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SENSORS

Arduino_MKRGPS

[ARDUINO](#) [LICENSE](#) LGPL-2.1 [V1.0.0](#) [Arduino](#) [25/10/2025](#)[Arduino](#) <info@arduino.cc><http://github.com/arduino-lib...> info@arduino.cc

Allows you to read the location from the GPS on your MKR GPS shield.

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The Arduino MKR ENV library allows you to read the information acquired through complex calculations by the Global Positioning System (GPS) receiver on the [MKR GPS Shield](#). This library stores and manages the NMEA strings generated by the GPS receiver so that you can get directly the relevant information you need from the GPS.

To use this library:

```
1 #include <Arduino_MKRGPS.h>
```

The GPS on the MKR GPS Shield can be managed using the Serial1 interface at 9600 8n1. On that serial port you receive the raw messages that can be read using the syntax specified by the [NMEA standard](#). You can manage the board with an I2C connection or through the header pins to an underlying MKR board. The library supports transparently each set up and you just need to specify the mode you are using with the proper `begin()` parameter.

Methods

begin()

Initialize the GPS.

Syntax

[Help](#)

```
1 GPS.begin()  
2 GPS.begin(mode)
```

Parameters

GPS_MODE_I2C

to use the MKR GPS with the I2C cable (default setting),

GPS_MODE_SHIELD

if using the MKR GPS as a shield.

Returns

1 on success, 0 on failure.

Example

```
1 if (!GPS.begin()) {  
2     Serial.println("Failed to i  
3     while(1);  
4 }
```

See also

end()
available()
latitude()
longitude()
speed()
course()
variation()
altitude()
satellites()
getTime()
standby()
wakeup()

▼ end()

De-initialize the GPS.

Syntax

```
1 GPS.end()
```

Parameters

None.

Returns

Nothing.

Example

```
1 if (!GPS.begin()) {  
2     Serial.println("Failed to i  
3     while(1);  
4 }  
5  
6 // Read GPS data here...  
7  
8 // Done working with the GPS  
9 GPS.end();
```

See also

[begin\(\)](#)
[available\(\)](#)
[latitude\(\)](#)
[longitude\(\)](#)
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> **available()**

▼ **latitude()**

Read the latitude of the GPS.

Syntax

```
1 GPS.latitude()
```

None.

Returns

GPS latitude in degrees.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS data
4     float latitude = GPS.getLatitude();
5     float longitude = GPS.getLongitude();
6
7     // ...
8
9     // Print GPS data
10    Serial.print("Location: ");
11    Serial.print(latitude, 7);
12    Serial.print(", ");
13    Serial.println(longitude, 7);
14 }
```

See also

[begin\(\)](#)
[end\(\)](#)
[available\(\)](#)
[longitude\(\)](#)
[speed\(\)](#)
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[variation\(\)](#)
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[standby\(\)](#)
[wakeup\(\)](#)

▼ [longitude\(\)](#)

Read the latitude of the GPS.

Syntax

```
1 GPS.latitude()
```

Parameters

None.

Returns

GPS longitude in degrees.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS data
4     float latitude = GPS.latitude();
5     float longitude = GPS.longitude();
6
7     // ...
8
9     // Print GPS data
10    Serial.print("Location: ");
11    Serial.print(latitude, 7);
12    Serial.print(", ");
13    Serial.println(longitude, 7);
14 }
```

See also

- [begin\(\)](#)
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- [available\(\)](#)
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- [getTime\(\)](#)
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- [wakeup\(\)](#)

Read the ground speed of the GPS.

Syntax

```
1 GPS.speed()
```

Parameters

None.

Returns

GPS ground speed in km/h.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS data
4     float speed = GPS.speed();
5
6     // ...
7
8     // Print GPS data
9     Serial.print("Ground speed: ");
10    Serial.print(speed);
11    Serial.println(" km/h");
12 }
```

See also

- `begin()`
- `end()`
- `available()`
- `latitude()`
- `longitude()`
- `course()`
- `variation()`
- `altitude()`
- `satellites()`
- `getTime()`
- `standby()`
- `wakeup()`

▼ `course()`

| Read the course of the GPS.

Syntax

```
1 GPS.course()
```

Parameters

None.

Returns

GPS course in degrees.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3   // Read GPS data
4   float course = GPS.course();
5
6   // ...
7
8   // Print GPS data
9   Serial.print("Course: ");
10  Serial.print(course);
11  Serial.println(" degrees")
12 }
```

See also

`begin()`
`end()`
`available()`
`latitude()`
`longitude()`
`speed()`
`variation()`
`altitude()`
`satellites()`
`getTime()`
`standby()`
`wakeup()`

▼ `variation()`

Read the magnetic variation of the GPS.

```
1 GPS.variation()
```

Parameters

None.

Returns

GPS magnetic variation in degrees.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS data
4     float variation = GPS.variation();
5
6     // ...
7
8     // Print GPS data
9     Serial.print("Variation: ");
10    Serial.print(variation);
11    Serial.println(" degrees");
12 }
```

See also

[begin\(\)](#)
[end\(\)](#)
[available\(\)](#)
[latitude\(\)](#)
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▼ altitude()

Read the altitude of the GPS.

Syntax


```
1 GPS.altitude()
```

Parameters

None.

Returns

GPS altitude in meters.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3   // Read GPS data
4   float altitude = GPS.altitude();
5
6   // ...
7
8   // Print GPS data
9   Serial.print("Altitude: ");
10  Serial.print(altitude);
11  Serial.println("m");
12 }
```

See also

`begin()`
`end()`
`available()`
`latitude()`
`longitude()`
`speed()`
`course()`
`variation()`
`satellites()`
`getTime()`
`standby()`
`wakeup()`

▼ `satellites()`

Read the number of satellites being tracked by the GPS.

Syntax

```
1 GPS.satellites()
```

Parameters

None.

Returns

The number of satellites being tracked by the GPS.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS values
4     int satellites = GPS.satellites()
5
6     // ...
7
8     // Print GPS data
9     Serial.print("Number of satellites: ");
10    Serial.println(satellites);
11 }
```

See also

[begin\(\)](#)
[end\(\)](#)
[available\(\)](#)
[latitude\(\)](#)
[longitude\(\)](#)
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▼ **satellites()**

Read the number of satellites being tracked by the GPS.

Syntax

```
1 GPS.satellites()
```

Parameters

None.

Returns

The number of satellites being tracked by the GPS.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS values
4     int satellites = GPS.satellites()
5
6     // ...
7
8     // Print GPS data
9     Serial.print("Number of satellites: ");
10    Serial.println(satellites)
11 }
```

See also

[begin\(\)](#)
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[latitude\(\)](#)
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[getTime\(\)](#)
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▼ [getTime\(\)](#)

Read the current epoch time from the GPS.

Syntax

Parameters

None.

Returns

The current epoch time from the GPS.

Example

```
1 // Check if there is new GPS data
2 if (GPS.available()) {
3     // Read GPS values
4     unsigned long epochTime = GPS.getEpochTime();
5
6     // ...
7
8     // Print GPS data
9     Serial.print("Epoch time: ");
10    Serial.println(epochTime);
11 }
```

See also

- `begin()`
- `end()`
- `available()`
- `latitude()`
- `longitude()`
- `speed()`
- `course()`
- `variation()`
- `altitude()`
- `satellites()`
- `standby()`
- `wakeup()`

▼ `standby()`

Put the GPS in standby mode.

Syntax

```
1 GPS.standby()
```

Parameters

None.

Returns

None.

Example

```
1 // Put the GPS in standby mode
2 Serial.println("Standby mode")
3 GPS.standby();
4
5 // Wait for 10 seconds
6 delay(10000);
7
8 // Wake up the GPS
9 Serial.println("Wakeup");
10 GPS.wakeup();
11
12 // Wait for new GPS data to be
13 Serial.print("Waiting new loca
14 while (!GPS.available());
15
16 // ...
```

See also

[begin\(\)](#)
[end\(\)](#)
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[latitude\(\)](#)
[longitude\(\)](#)
[speed\(\)](#)
[course\(\)](#)
[variation\(\)](#)
[altitude\(\)](#)
[satellites\(\)](#)
[getTime\(\)](#)
[wakeup\(\)](#)

wakeup()
Wake up the GPS from standby mode.

Syntax

```
1 GPS.wakeup()
```

Parameters

None.

Returns

None.

Example

```
1 // Put the GPS in standby mode
2 Serial.println("Standby mode")
3 GPS.standby();
4
5 // Wait for 10 seconds
6 delay(10000);
7
8 // Wake up the GPS
9 Serial.println("Wakeup");
10 GPS.wakeup();
11
12 // Wait for new GPS data to be
13 Serial.print("Waiting new loca
14 while (!GPS.available());
15
16 // ...
```

See also

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